3SAT $\longrightarrow$ zero-finding with constant Jacobian

CLAIM. Given a 3CNF formula ϕ, construct a degree-7 map K_ϕ with

$$K_\phi^{-1}(0) \neq \varnothing \iff \phi \in \mathrm{SAT}, \qquad \det JK_\phi \equiv (-3456)^{n+m}.$$

1. **The gadget.** Fix $H : \mathbb{R}^3 \to \mathbb{R}^3$ with $\det JH \equiv -3456$, $H^{-1}(0) = \{r_0, r_+, r_-\}$, and one omitted target $\Delta \notin \mathrm{im}(H)$. The decoder $\beta(x, y, z) = x^2$ reads r_0 as 0 and $r_\pm$ as 1.

2. **The switch.** Define $T_s(v) = H(v) - (1-s)\Delta$. Solving $T_s(v) = 0$ is the same as asking H to output $(1-s)\Delta$. If $s = 1$, this asks for 0, which is possible. If $s = 0$, it asks for Δ, which is impossible. Thus

$$\exists v : \ T_s(v) = 0 \iff s = 1 \qquad (s \in \{0, 1\}).$$

3. **One clause and the stack.** For example, for $C_j = X_a \vee \neg X_b \vee X_c$, set

$$s_j = 1 - (1 - \beta(w_a))\,\beta(w_b)\,(1 - \beta(w_c)).$$

This is 1 exactly when the clause is true. Attach its block $T_{s_j}(v_j) = H(v_j) - (1 - s_j)\Delta$. The complete map is the stack

$$K_\phi = \Big(H(w_1), \ldots, H(w_n), T_{s_1}(v_1), \ldots, T_{s_m}(v_m)\Big).$$

Each clause block can vanish exactly when its clause is true, so $K_\phi = 0$ exactly when the encoded assignment satisfies every clause.

4. **The Jacobian.** Changing s_j only translates the output of H; it does not change the derivative with respect to the fresh block v_j. The full Jacobian is block lower triangular with $n + m$ copies of JH on its diagonal, so $\det JK_\phi = (-3456)^{n+m}$ and $\deg K_\phi \leq 7$.